

Phase 7 Acceptance Record: Pass One Local Speech-to-Text (ASR)

Document ID: CAW-EV-P7-01
Date: September 2026
System: Case Ace v2.0
Phase: 7 - Pass One, Local ASR
Status: PASSED & VERIFIED

1. Scope & Acceptance Criteria Verification Matrix

Phase 7 implements Pass One Local Speech-to-Text executed 100% within the browser. Its sole purpose is to locate candidate identifier tokens for local acoustic redaction (Pass Two) without transmitting audio to external networks.

2. Automated Test Execution Evidence

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CASE ACE v2.0 - PHASE 7: PASS ONE LOCAL ASR VERIFICATION
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--- TEST SUITE 1: Voice Activity Detection & Segmentation ---
  ' [PASS] detectSpeechSegments segments audio into distinct conversational turns
  ' [PASS] inferSegmentSpeaker differentiates adviser vs client acoustic characteristics

--- TEST SUITE 2: Word-Level Timestamps & Confidence Scores ---
  ' [PASS] processAsrInference produces word-level timestamps and valid confidence scores

--- TEST SUITE 3: Low-Confidence Escalation Policy (<0.70) ---
  ' [PASS] Low confidence words (<0.70) are flagged and escalated rather than discarded

--- TEST SUITE 4: Real-Time Progress & Dynamic ETA ---
  ' [PASS] Progress updates emit monotonically increasing percentage and ETA

--- TEST SUITE 5: Speaker Attribution & Channel Disambiguation ---
  ' [PASS] Preserves exact Webex telephony stereo speaker channel split

--- TEST SUITE 6: Volatile Memory Hygiene & Destruction ---
  ' [PASS] VolatileSessionStore stores local ASR result and wipes it cleanly on destruction

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PHASE 7 VERIFICATION RESULTS: 7/7 TESTS PASSED
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3. Architecture & Documentation Deliverables

Architecture Guide: [docs/local-asr-architecture.md](file:///Users/mothership2/Library/CloudStorage/GoogleDrive-admin@adviceintelligence.tech/My%20Drive/Google%20AI%20Studio/case-ace-v2/docs/local-asr-architecture.md)
Engine Implementation: [client/src/asr/localAsrEngine.ts](file:///Users/mothership2/Library/CloudStorage/GoogleDrive-admin@adviceintelligence.tech/My%20Drive/Google%20AI%20Studio/case-ace-v2/client/src/asr/localAsrEngine.ts)
Sandboxed Worker: [client/src/workers/localAsrWorker.ts](file:///Users/mothership2/Library/CloudStorage/GoogleDrive-admin@adviceintelligence.tech/My%20Drive/Google%20AI%20Studio/case-ace-v2/client/src/workers/localAsrWorker.ts)
UI Components: [client/src/components/HardwareBenchmarkBanner.tsx](file:///Users/mothership2/Library/CloudStorage/GoogleDrive-admin@adviceintelligence.tech/My%20Drive/Google%20AI%20Studio/case-ace-v2/client/src/components/HardwareBenchmarkBanner.tsx) & [client/src/components/LocalAsrProgressModal.tsx](file:///Users/mothership2/Library/CloudStorage/GoogleDrive-admin@adviceintelligence.tech/My%20Drive/Google%20AI%20Studio/case-ace-v2/client/src/components/LocalAsrProgressModal.tsx)
Acceptance Test Suite: [scripts/test-phase7.mjs](file:///Users/mothership2/Library/CloudStorage/GoogleDrive-admin@adviceintelligence.tech/My%20Drive/Google%20AI%20Studio/case-ace-v2/scripts/test-phase7.mjs) & [test/unit/localAsr.test.ts](file:///Users/mothership2/Library/CloudStorage/GoogleDrive-admin@adviceintelligence.tech/My%20Drive/Google%20AI%20Studio/case-ace-v2/test/unit/localAsr.test.ts)